

INDIAN EXPRESS UPSC IAS EDITION HD 07~06~2026
-:FOR UPSC IAS ASPIRANTS:-
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Can't So Then You Have To Increase Your Efforts
All the topics of this UPSC IAS Edition are directly or
indirectly important for the prelims & main examination.
There are some topics which can be coded in answer
writing of other topics in the main exam.

US adds \$38 mn for Ebola as CDC warns outbreak could match 2014

Reuters
Washington, June 6

THE UNITED States said on Friday it would provide an additional \$38 million for Ebola response efforts, as the CDC warned that without strong public health interventions the outbreak in the Democratic Republic of Congo could match or surpass the 2014-2016 West Africa crisis.

The additional funds bring the total committed by the US to more than \$200 million in direct funding, the State Department said in a statement.

The department, which did not detail how the money would be spent, said it was working closely with the US Centers for



A health worker disinfects an ambulance at the Mongwalu treatment in Congo on Saturday. AP

Disease Control and Prevention, as well as the DRC and Uganda to “mount a rapid and comprehensive response” to the outbreak. The CDC published three official scientific reports

on the outbreak on Friday, in part to marshal resources across the international community, said Dr. Satish Pillai, the incident manager for the agency's Ebola response.

The documents are the first such reports to be published since the World Health Organization declared an emergency of international concern on May 17 over the Bundibugyo strain of Ebola, for which there is no approved vaccine or treatment.

CDC model scenarios show that without strong public health interventions, the outbreak of the Bundibugyo strain of Ebola in the DRC could become as large as, or even larger than, the 2014-2016 West Africa outbreak, the agency said.

CHINA'S HUMANOID ROBOT MAKERS SAY THEY HAVE THOUSANDS OF ORDERS, BUT EXPERTS WARN DEMAND STILL TRAILS THE COUNTRY'S PRODUCTION CAPACITY

China can build humanoids at scale; the hard part is finding enough buyers

Chan Ho-Him
Hong Kong, June 6

CHINESE-MADE humanoid robots are making waves with their ability to do backflips, direct traffic, and even make coffee as the companies developing them seek ways to expand and dominate the market.

Robot makers in China say they have thousands of orders from both the government and private businesses for humanoids that can do such things as sort parcels at postal centers, as the country finds ways to cope with an aging population and rising labor costs.

However, some experts believe demand for humanoids

lags the capacity to build them.

China and the United States dominate research for what Morgan Stanley estimates is a \$5 trillion humanoid robots market. By some measures, the US holds an upper hand in developing the artificial intelligence for such robots' high-level computing power, or “brains.” But as the world's factory floor, China leads in mass production capacity, supplies of hardware and harvesting of data for training robots.

The Shanghai-based startup Matrix Robotics makes humanoid robots that employ AI. Its flagship humanoid robot, the “MATRIX-3,” stands nearly 5.6 feet (1.7 meters) tall and is

equipped with hands able to make finely controlled movements. They are priced at around \$99,000 per unit.

Customers for the roughly 1,000 orders it has received include coffee chains and hotels, its founder and CEO Allan Zhang, who formerly worked for Tesla, said at a recent robotics expo in Macao.

So far, Matrix has made only a few hundred of the robots, though it said it will be capable of delivering 5,000 units within this year, depending on the number of orders.

EngineAI, a startup based in southern China's Shenzhen, says its full-sized humanoid robots could be used as security



Workers assemble a robot at the LY iTech Beijing Super Factory in the Beijing E-Town on the outskirts of the city. FILE

guards and museum guides. They also perform, with dancing and boxing.

A basic edition of its humanoid costs 180,000 yuan (\$26,600). “The next step will be

to move into more real-life scenarios,” said Issac Li, EngineAI's head of brand and marketing.

Most humanoid robots are still performative rather than functional, falling short of working in messy, unpredictable environments, said Sam Sacks, a senior fellow at the New America think tank focused on Chinese technology.

“The use cases of these robots are still so limited,” said Chibo Tang of the venture capital firm Gobi Partners, which invests in technology startups including robotics companies. “Without the demand and without that scale from the market, these companies are not able to really go into mass production.”

China had more than 140 humanoid robot manufacturers and more than 330 models in 2025, according to the Ministry of Industry and Information Technology. Last year, the Chinese government even publicly warned about the risk of a bubble in the industry given the lagging state of commercialization and applications.

Corporate and academic labs are buying humanoid robots for research. And in China, many of the more than 2 billion yuan (\$295 million) worth of orders in 2025 came from state-owned enterprises for use in places such as power plants, data centers or for entertainment, Morgan Stanley said.

“The economics are tough: humanoid robots remain expensive to produce, fragile in operation, and dependent on highly structured environments to function,” Sacks explained. There's “a long way to go to get to a level of functionality where people will actually feel comfortable having them in their homes providing care for elderly or children,” she said.

The more viable commercial path will more likely be through industrial and logistics settings, Sacks said. But many factories in China and elsewhere already are equipped with non-humanoid robotic arms that perform repetitive functions and may not need humanoid robots. AP



NEW DELHI,
JUNE 7, 2026

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The Sunday EXPRESS



Prime Minister Narendra Modi chairs a meeting of the Economic Advisory Council in New Delhi, Saturday. PTI

Sense in PM-EAC meeting: Moves to attract foreign funds can get \$70 billion

Siddharth Upasani
New Delhi, June 6

THE STEPS taken by the government and the Reserve Bank of India on Friday are likely to attract around \$70 billion in foreign funds, according to the discussions at the meeting of the Economic Advisory Council chaired by Prime Minister Narendra Modi on Saturday.

While the Ministry of Finance did away with the tax on long-term and short-term capital gains as well as the withholding tax on investment by FIIs in government bonds, the RBI made it easier for banks to mobilise foreign deposits and for PSUs to raise external commercial borrowings.

The inflow of a sizable por-

tion of foreign funds is contingent on the inclusion of government bonds in the Bloomberg Global Aggregate Bond Index. This is crucial to attracting passive investment flows; several global funds track the weight a certain country has in these bond indices, and invest accordingly. The inflows into the government debt market helps lower the interest paid by the Centre on its borrowings.

After the meeting, Prime Minister Narendra Modi, who chaired the meeting said on social media platform X, "Deliberated on a wide range of issues relating to India's economic transformation and long-term development priorities. Also shared perspectives on adding

»CONTINUED ON PAGE 2

PROJECT GLASSWING

Key cybersecurity agencies set to get access to Anthropic's Mythos

Discussions on to provide AI model to largest IT services companies, too

Soumyarendra Barik
New Delhi, June 6

US ARTIFICIAL intelligence company Anthropic is extending access to its closely guarded cybersecurity initiative, Project Glasswing, to a select set of organisations in India, including

key government entities that are responsible for safeguarding cybersecurity across sectors like banking, telecom, and power, *The Indian Express* has learned.

Government agencies such as the Indian Cyber Crime Coordination Centre (I4C); Indian

»CONTINUED ON PAGE 2

E. EXPLAINED

Firewalling systems

AI model Mythos can detect critical software vulnerabilities. This matters for India as it shields vital infrastructure like banking and power, even as Anthropic battles the US Pentagon over boundaries of AI misuse.

Mythos

Computer Emergency Response Team (CERT-In); National Critical Information Infrastructure Protection Centre (NCIIPC); and the Department of Telecommunications' Digital Intelligence Platform (DIP) are among the state-owned entities that are understood to have received access to Anthropic's advanced frontier AI model Mythos.

Anthropic is also said to have extended access to some cybersecurity-focused research institutions in India, while discussions are underway on providing Mythos Preview to dedicated cybersecurity and AI teams within some of the country's largest IT services companies, a senior government official said on condition of anonymity.

This move comes as part of Anthropic's announcement that it would expand Project Glasswing beyond its initial cohort of participants in the US and UK to more than 15 countries, including India.

Under the programme, vetted organisations are granted access to Claude Mythos Preview, an unreleased AI model that Anthropic says is capable of identifying critical software vulnerabilities at a level that could fundamentally alter the balance between cyber attackers and defenders. The company has so far restricted access to the model, citing concerns that its capabilities

could be misused if released publicly.

The official added that NCIIPC, which falls under the National Security Advisor in the Prime Minister's Office, and CERT-In had requested access to Mythos to identify vulnerabilities within India's critical infrastructure such as banking and power. *The Indian Express* had earlier reported that the Indian government was in conversations with Anthropic's senior leadership in the US for assistance in gaining access to the model.

Anthropic did not respond to a request for comment.

"Following several weeks of close collaboration with our Project Glasswing partners, the security industry, open-source software maintainers, and the US government, we're extending the partnership to approximately 150 new organisations. Each one will need to meet our security requirements before they gain access... The organisations in this new group are based in more than 15 countries, and most provide critical infrastructure to many more," Anthropic said in a blog post.

The new entities with which it is planning to share Mythos Preview operate in industries such as power, water, healthcare, communications, and hardware, Anthropic said. And many of the new partners are vendors — companies or non-profits that maintain codebases that are relied upon by lots of other organisations around the world, including governments.

PM-EAC

more momentum to the reforms journey and ensuring 'Ease of Living' as well as 'Ease of Doing Business'."

According to a source, who did not wish to be named, while there wasn't much concern around the West Asia crisis, a key discussion point revolved around the impact of El Nino and sub-par rainfall as forecast by the IMD. "How can India reduce its vulnerability due to the vagaries of the monsoon — this was also discussed," the source said.

Economists are of the opinion that Friday's decision to scrap the short-term and long-term capital gains tax on investments by FIIs in government bonds as well as the withholding tax they must pay on their interest income from these debt instruments, will likely accelerate the inclusion of Indian sovereign debt in the Bloomberg Global Aggregate Bond Index. This alone could result in \$20-25 billion inflows over the 10 months.

In recent years, government debt — the bonds it issues to borrow money from the market to cover its fiscal deficit, or the difference between its income and expenditure — has been added to multiple global bond indices. This began with JPMorgan adding India to its emerging

market index beginning June 2024. Subsequently, Bloomberg included India in its Emerging Market Local Currency Index from January 2025 and FTSE Russell to a similar one from September 2025.

"Bloomberg index inclusion, I believe, is on the verge of happening," a bond market participant told *The Indian Express*.

In mid-January, Bloomberg Index Services Ltd (BISL) had deferred the inclusion of Indian government bonds into its flagship global index, saying it will provide another update by mid-2026. "Overall, responses indicated broad support for the long-term trajectory of the Indian government bond market and for its potential eventual inclusion in global investment grade benchmarks," BISL had said in a statement on January 13. "At the same time, a number of respondents highlighted important operational and market infrastructure considerations that merit further evaluation..."

In addition to the removal of taxes on FIIs, Friday also saw the RBI announce several measures to boost foreign inflows, including a temporary concessional forex swap facility for PSUs for their External Commercial Borrowings and the revival of the wildly successful Foreign Currency Non-Resident (Bank) Deposits scheme of 2013, with the central bank bearing the full exchange rate hedging cost.



Verify nationality of 2,860 people for their deportation: India to Bangladesh

Shubhajit Roy
New Delhi, June 6

AS THE bordering states of West Bengal and Assam intensify their crackdown on illegal immigrants from Bangladesh, the Indian government has requested the neighbouring country to verify the nationality of 2,860 people so that they can be sent back.

This assumes significance at a time when hundreds of Bangladeshi immigrants, who are reported to have entered India illegally, have flocked to the borders in the recent weeks, as reported by *The Indian Express*. This movement of people has been fuelled by the announcement of the newly elected BJP government in West Bengal, led by Chief Minister Suvendu Adhikari, that it will follow the “detect, delete and deport” policy; the government has so far detained about 400 undocu-

mented Bangladeshi immigrants in its detention centres.

This is also an outcome of the Special Intensive Revision (SIR) exercise in the state, which has led to a panic among the immigrants in the border state.

Ministry of External Affairs spokesperson Randhir Jaiswal said, “We have requested Bangladesh to verify the nationality of over 2,860 people whom we think are Bangladeshi nationals, and they are staying here in India. We are yet to get a response on this matter.” Addressing the weekly media briefing Friday, Jaiswal said: “All foreign nationals in India, if they are here illegally, including from Bangladesh... we have laws to deal with them, and they’ll be dealt with accordingly.”

“As far as deportation of these people is concerned, there is a bilateral mechanism in place... We refer these cases to the Bangladeshi side for them to verify the

E. EXPLAINED

The Diplomatic Challenge

The diplomatic challenge for India is to navigate the issue of deporting illegal Bangladeshi nationals as the BJP has won the two bordering states of West Bengal and Assam on the poll plank of deporting illegal immigration from the neighbouring country. The Bangladesh border guards have not been taking the immigrants back since they too need to verify their nationality. Dhaka, meanwhile, has expressed concern over “push in attempts”, insisting that any return should be through formal, legal and diplomatic channels.

nationality of these people, and once it is verified, we take forward the deportation process.”

“We have several or many of these requests, which are still pending with the Bangladeshi side, and we hope and expect that they’ll be expedited as soon as possible so that the deportation of people who are illegally staying in India can be taken forward in a smooth and efficient manner,” Jaiswal said.

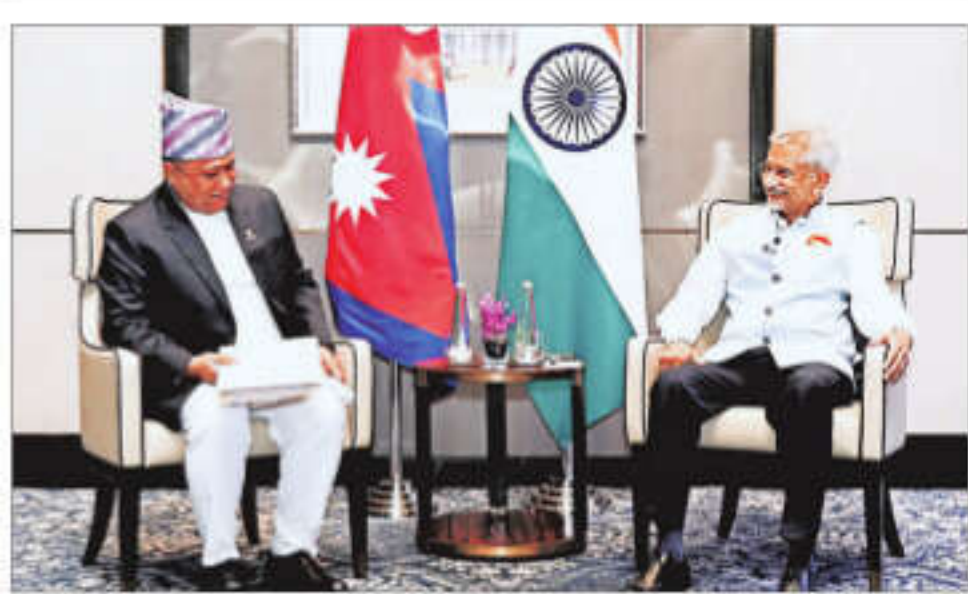
Officials said they have been raising this issue with the Bangladeshi side for the past several years. In early May, Delhi had raised the issue of illegal immigrants with Dhaka and expressed hope that it will expedite nationality verification for smooth repatriation. The development came days after the BJP swept the West Bengal Assembly elections.

A day after the poll results, Bangladesh Foreign Minister Khalilur Rahman was quoted in

the ruling Bangladesh Nationalist Party’s official Facebook page as saying that Dhaka will take action if “push-in” incidents occur amid the change of power in the bordering state.

Officials said the issue of illegal migration from Bangladesh to India is being discussed at several bilateral meetings between the two countries, including at the home secretary level.

India and Bangladesh have put in place mechanisms, including a Coordinated Border Management Plan, to address problems arising from illegal border crossing, reduce the scope for incidents on the border and enhance cooperation between the borderguarding forces of the two countries. The two sides are also cooperating in undertaking fencing of the vulnerable patches with a view to prevent criminal activities and illegal movements along the border, officials said.



External Affairs Minister S Jaishankar and Nepal Foreign Minister Shishir Khanal during a meeting in New Delhi on Saturday. PTI

India, Nepal vow to take bilateral ties to new heights

Shubhajit Roy
New Delhi, June 6

INDIA AND Nepal completed the internal processes to enforce the mutual legal assistance pact to tackle “cross-border crimes”, as the visiting Foreign Minister of Nepal Shishir Khanal met External Affairs minister S Jaishankar on Saturday.

The two sides also made progress in the sectors of health, UPI, innovation and startups. Khanal’s visit comes days after the chief of Nepal’s ruling RSP, Rabi Lamichhane, met the Indian leadership in Delhi. After the bilateral meeting, Jaishankar posted on X, “A pleasure to meet FM Shishir Khanal of Nepal and his delegation today. A detailed discussion on our unique partnership, including on development cooperation, connectivity, energy, especially hydropower development, education, health, capacity building, digital, culture and sports.”

The MEA released an official statement: “Both Ministers welcomed the completion of internal processes for entry into force of the India-Nepal Mutual Legal Assistance Agreement in Criminal Matters (MLAA). This Agreement will benefit the people of India and Nepal by providing an institutional legal framework to enhance the effectiveness of investigation, prosecution and judicial proceedings relating to cross-border crimes.” The min-

isters vowed to take the bilateral ties to “new heights” while reviewing the full spectrum of the relations, covering development cooperation, connectivity, trade and transit, energy, and people-to-people ties, the MEA said.

The Nepalese Foreign Minister also met National Security Advisor Ajit Doval. After the bilateral talks, Jaishankar handed over 72 health facilities and 12 cultural heritage sector projects, completed under India’s post-2015 ‘Earthquake Reconstruction Assistance’ to Nepal, to the visiting Foreign Minister. The ministers jointly launched the Peer-to-Peer (P2P) linkage between Unified Payments Interface (UPI) of India and National Payments Interface (NPI) of Nepal for facilitating cross-border personal remittances.

The MEA said the ministers witnessed the signing of MoU between Digital India Bhashini and Kathmandu University for co-creating National Digital Infrastructure for “Voice First” Language Translation platform. This is the first official visit by the Nepalese Foreign minister after the new government led by PM Balendra Shah assumed office in March this year.

“Nepal is a priority partner of India under its ‘Neighbourhood First’ policy and the visit reinforced the tradition of regular high-level exchanges between the two close and friendly neighbours...,” the MEA said.

Harikishan Sharma
New Delhi, June 6

A GOVERNMENT-COMMISSIONED evaluation of central rural sector schemes, including the Deendayal Antyodaya Yojana-National Rural Livelihoods Mission (DAY-NRLM), has found that most self-help groups (SHGs) for women did not start income-generating businesses with the funds provided.

Instead, the revolving fund of Rs 20,000-Rs 30,000 offered per group as part of the Centre’s flagship programme to involve poor rural women into SHGs and train them for small enterprises was often spent on household expenses, according to the report submitted to the Rural Development Ministry.

The study is significant given that the Finance Ministry last year directed that no centrally funded scheme will be extended beyond March 31, 2026, unless it undergoes a third-party evaluation. This review will therefore decide whether and how the DAY-NRLM will continue in the 16th Finance Commission cycle beginning in April 2026.

The evaluation was undertaken by Sambodhi Research and Communication Pvt Ltd, a think tank engaged by NITI Aayog’s Development Monitoring and Evaluation Office (DMEO) for the purpose.

Sambodhi, in an email to *The Indian Express*, said the official DAY-NRLM/RBI guidelines specify that “SHG-linked financial support may serve multiple purposes, including social needs, housing repair, sanitation, healthcare-related expenditure and other activities that contribute to household resilience”.

The report titled ‘Evaluation of Centrally Sponsored Schemes in Package 5-Rural Development Sector’ also flags that there was pressure on officials to form SHGs quickly to meet targets, even in villages where business opportunities

Officials pressured to form SHGs for women quickly under DAY-NRLM scheme to meet targets, says report



Those SHG members who took bank loans often used them for building houses, toilets, medical purposes, says the report. FILE

were limited.

“Only a small proportion of SHGs have initiated group-level enterprises, and nearly half did so without accessing loans. This is primarily driven by the pressure to form SHGs after the PIP [Participatory Identification of the Poor] exercise to meet targets, despite a significant disconnect between SHG formation and the availability of viable business opportunities in rural areas,” the report notes.

According to the report, many SHG members are reluctant to take loans because there is no assured market for the products made through group enterprises.

“Due to this uncertainty, they fear being unable to repay the loan, discouraging them from borrowing,” says the report. As a result, “the RF (Revolving Fund) is often equally distributed among members and utilised to meet their immediate consumption needs.”

Those SHG members who took bank loans often used them for building houses, toilets, medical purposes etc,

the report shows.

Access to other financial support under the scheme was also low: fewer than half the respondents received the Community Investment Fund (CIF), only 42% accessed bank loans, and less than 4% received the Community Enterprise Fund (CEF) meant for larger ventures.

While access to RF was reported by 74.6% of respondents in 13 states, just 49.9% of respondents reported receiving the CIF, 41.8% had access to bank loans, and only 3.8% reported access to the CEF.

“This was primarily due to limited on-ground support from VOs [village organisations], CRPs [community resource persons], and CLFs [cluster-level federations] coupled with the fact that only 15% of the SHGs had initiated group enterprise activity,” noted the report.

“The uptake of bank loans/cash credit limit was even lower due to fear of loan defaults, as SHG members perceived weak market demand for their products,” the report

states. The evaluation exercise involved a survey of 2,206 households across 11 states (Andhra Pradesh, Bihar, Chhattisgarh, Maharashtra, Rajasthan, Tripura, Arunachal Pradesh, Himachal Pradesh, Kerala, Odisha, Uttar Pradesh and two Union Territories (Jammu and Kashmir, and Dadra and Nagar Haveli and Daman and Diu).

Of the 2,206 households surveyed, about 86% (1,895 households) availed benefits of NRLM. Launched in 2011 by the UPA government as the National Rural Livelihood Mission and renamed in 2016, DAY-NRLM aims to reduce rural poverty by mobilising women into SHGs and linking them with banks and government support.

Apart from the revolving fund at the rate of Rs 20,000-30,000 per SHG, groups can get up to Rs 2.5 lakh as a Community Investment Fund to support their income-generating and livelihood activities.

According to government data, more than 10 crore rural women have been mobilised into over 90 lakh SHGs, which together have received nearly Rs 59,000 crore in capital support since the scheme began.

Reacting to an online version of this story, Abhishek Sharma, deputy vice-president, Sambodhi, said: “The official DAY-NRLM/RBI guidelines do not treat all non-enterprise use of SHG-linked credit as misuse. The programme framework recognises that SHG-linked financial support may serve multiple purposes, including social needs, high-cost debt swapping, housing repair or construction, sanitation, healthcare-related expenditure, livelihood stabilisation, and other activities that contribute to household resilience and financial inclusion.”

“We therefore wish to place on record that Sambodhi does not share the interpretation... and overall framing of the article,” he said in an email.

Let us talk about education



FIFTH COLUMN
BY TAVLEEN SINGH

IT IS astonishing that it has become fashionable to talk about education. For the wrong reasons maybe, but at least we are discussing the most undiscussed and, in my view, the most vital reform that India needs. So far, all talk has been about NEET and leaked papers and the venality that has been revealed in the CBSE exam system. Desperate young students have been driven to suicide because of these corrupt practices. This alone should shame those responsible. So, when the noise and drama of these events die, please, please let us continue talking about the rot in our education system.

Maybe this time we will talk about it a little longer because of India's newest political party making leaked papers their first cause. By the time you read this, the leader of the Cockroach Janata Party should have landed in Delhi and probably been arrested before he manages to hold a protest rally at Jantar Mantar. It is my fervent hope that even if the protest fails, the cause that he claims has brought him back home will not be forgotten. There must be accountability and heads must roll. He has demanded the resignation of Dharmendra Pradhan, and if this man had any self-respect, he would have already resigned, but this will solve nothing.

The rot goes deep and has spread from the top to the very bottom. The reason why our political leaders and high officials never send their children to government schools is because they know that they have built schools that offer not real learning but barest literacy. When Indian children from Tamil Nadu and Himachal competed in the Programme for International Student Assessment (PISA) in 2009, they came second last. In this exam, 15-year-olds were tested for their skills in reading, mathematics and science. Alarm bells should have gone off across India but if our babus have one superpower, it is knowing how to keep bad news from reaching the ears of their political masters.

We did not need an international test to establish that schools run by the Indian state are abysmal. If our prime ministers have had one thing in common, it is their neglect of education. Jawaharlal Nehru built the IITs but appeared not to notice that the only children able to get into them were from upper caste, middle class families. He also did not notice that on his watch was created a higher education system that was so tightly controlled by the government that it killed all chances of excellence or research. One reason is meddlesome officials. If there is to be reform, then it should start with the abolition of the University Grants Commission. It is an unneeded busybody that gives officials unneeded power.

Before writing this piece, I had a chat with Michel Danino who knows from the inside how the system works. He worked in the National Council of Educational Research and Training (NCERT) and recently got into trouble with the Supreme Court for telling the truth about corruption in the judiciary in a school textbook. When I asked him what were the three most important reforms that needed to be made in the school system, he said that for a start, instead of students being treated as a 'passive sponge', they need to be encouraged to participate. In other words, less rote learning.

The second reform he recommends is the setting up of proper teacher training institutes. And third, that there be investment in school infrastructure. Government schools lack such basic things as libraries, playgrounds, clean water and toilets. I have visited village schools in many states and seen some that are just empty sheds. So, I could not agree more. The irony is that the one reform that has been made is for teachers to be paid hefty salaries. These are sometimes ten times higher than the paltry salaries teachers earn in the thousands of private schools that have sprouted across rural India.

It is heartbreaking to think of how disadvantaged Indian children will be when they go out into a world where the competition could be AI children. Already, we have 'graduates' who cannot find jobs because they have left college without being able to articulate an idea or solve a simple equation. It has been said more than once that the problem in India is not unemployment but unemployability.

In 2013, when I was a passionate Modi-*bhakt* and hoped with all my heart that he would be able to bring the changes India so desperately needs, I was once summoned to meet him. He was still Chief Minister of Gujarat but knew in his heart that when the general election came the following year, he would be prime minister. He asked me what I felt were the things that most urgently needed to be done. And, I said he needed to rectify the mistakes that Congress prime ministers had made, which was to neglect education and healthcare.

We talked uninterrupted for more than an hour at the end, of which I was convinced that I had made my point. Clearly, I had not or by now we should have seen a revolutionary improvement in government schools that are in states governed by BJP chief ministers. Today that includes all major Hindi-belt states, Maharashtra and now West Bengal. Will the Cockroach noises bring some urgency? It is my sincerest hope that this will happen.

How an IMD discovery in 1920s changed monsoon forecasts



HISTORY HEADLINE
BY AMITABH SINHA

THE EL NIÑO phenomenon is fairly well known in India, thanks to widespread discussion in the media in connection with its impact on the Indian monsoon. El Niño Southern Oscillation, or ENSO, is a large-scale ocean-atmosphere interaction playing out in the Pacific Ocean region that happens to be the biggest cause of year-on-year variations in global climate. It has profound impacts on weather events everywhere, including the Indian monsoon.

What is much less known is that this global phenomenon, at least the atmospheric component of this ocean-atmosphere interaction, was first discovered by the India Meteorological Department (IMD) in the early 1920s. It was British mathematician Sir Gilbert Walker, working as head of IMD, a position that was at that time described as Director General of Observatories, who had noticed the see-saw behaviour of air pressure between the east and west Pacific Ocean, something that became known as the Southern Oscillation, or SO.

While working to develop a reliable forecast model for the Indian monsoon, Walker, who had a nearly 21-year tenure at IMD from 1904 to 1924, had decided to look at external influences, having realised that the monsoon was not an isolated phenomenon over the Indian region. Meticulously going through all the weather observation data that used to be collected at the time, Walker noticed three large systems, two in the Pacific and one in the Atlantic Ocean, in which the atmospheric pressures periodically oscillated between being very high



Sir Gilbert Walker, who headed IMD, found that the Southern Oscillation could have an impact on the Indian monsoon

and very low at the opposite ends. The Southern Oscillation was so named because it was relatively south of the other two, which were called the North Atlantic Oscillation and North Pacific Oscillation.

He found the Southern Oscillation to be the most persistent and also the most likely to have an impact on the Indian monsoon.

Decades later, Norwegian-American meteorologist Jacob Bjerknes linked the Southern Oscillation to the periodic warming and cooling of the equatorial Pacific Ocean, or El Niño. Natives of the western coastline of South America, in the countries of Peru and Ecuador, were aware of this periodic change in sea surface temperatures for centuries. But it was Bjerknes who showed in the late 1960s that the two processes, the ocean phenomenon of El Niño and the atmospheric phenomenon of Southern Oscillation, formed a feedback loop in which one reinforced the other.

This discovery gave the scientists their first understanding of this complete process. El Niño and La Niña, the two opposite phases of ENSO, are now known to in-

fluence every large-scale weather phenomenon in the world. In the El Niño phase, the equatorial Pacific Ocean, off the coast of South America, gets warmer than normal, while in the La Niña phase, it gets cooler. El Niño has a general warming impact on the planet. It is known to suppress rainfall over the Indian and Australian region, including southeast Asia, but causes bountiful rains to happen over the North Americas. La Niña has opposing impacts.

Walker's discovery of the southern oscillation marked a major improvement in the scientific understanding of monsoon and resulted in greater scientific grounding of its forecasts. Before that, forecasts, issued since the establishment of IMD in the 1870s, relied heavily on statistical correlations.

However, monsoon forecasts did not suddenly become accurate. Even decades after discovering the ENSO link, IMD struggled to provide reliable forecasts. That is because the monsoon is a highly complex system, and its relationship with ENSO is not causal. In fact, because of the lack of reliable predictors, the IMD, for a very long time, between 1932 and 1988, had stopped providing monsoon forecasts for the entire country. Its forecasts used to be limited only to northwest India and the peninsular region. And even this was not quantitative. Instead of mentioning the percentage of normal rainfall expected during the season, as it does now, IMD then used terms like "slightly above normal" or "not far from normal".

It was only in 1988 that IMD restarted all-India forecasts and reintroduced the quantitative element into its predictions. Over the years, IMD has significantly upgraded its computational abilities, observational network, and prediction models, resulting in remarkable improvements in long-range forecasts. But new challenges have arisen, particularly those related to extreme weather events.

The writer is Deputy Editor, Climate & Science, The Indian Express

SUPPLY SECURITY PRIORITISED OVER PRICE CONCERNS AMID HIGH GAS DEMAND

US tops India’s LNG basket in May, imports recover to pre-war levels

Sukalp Sharma
New Delhi, June 6

WITH THE heavy disruption in liquefied natural gas (LNG) imports from key suppliers Qatar and the UAE, the US emerged as India’s largest source of super-chilled gas in May, followed by Nigeria, Oman, and Angola, as New Delhi’s LNG imports recovered to pre-war levels despite the surge in international LNG prices.

While some major LNG importing countries reduced imports amid sky-high prices, India prioritised supplies over price amid demand from various sectors, which include city gas distribution, fertiliser, power, and ceramics.

The effective closure of the Strait of Hormuz has upended the regular flow of LNG to India from key suppliers Qatar and the UAE, leading to a scramble for LNG cargoes from alternate geographies. The supply disruption has not been as severe as initially anticipated due to a combination of alternative suppliers raising exports and demand curtailment in various parts of the world. India depends on LNG imports to meet about half of its natural gas requirement, and about 60% of those imports came through the critical maritime chokepoint of the Strait of Hormuz, primarily from Qatar, and also the UAE. The two Gulf



India depends on LNG imports to meet about half of its natural gas requirement. REUTERS

nations are major LNG exporters. Ship tracking data from commodity market analytics firm Kpler shows that India imported 2.2 million tonnes of LNG in May, 13.5% higher over April, and 32% over March, which was the first month impacted by the West Asia war and the consequent closure of the Strait of Hormuz. LNG imports in May were, in fact, 6% higher over the corresponding month of last year, and 6.4% higher than the 2025 monthly average.

As no LNG volumes came to Indian shores from Qatar for the second month in a row, and just 0.1 million tonnes came from the UAE, LNG importers continued with their pivot to alternative suppliers—that don’t depend on the Strait of Hormuz—to secure emergency spot cargoes.

Volumes from the US tripled to 0.9 million tonnes, accounting for about 41% of India’s overall LNG imports for the month, as per Kpler data. In 2025, the average monthly LNG imports from the US were 0.2 million tonnes.

Nigeria was in the second spot with 0.5 million tonnes—flat on a month-on-month basis—followed by Oman and Angola at 0.3 million tonnes each. In April, LNG imports from Oman were 0.6 million tonnes, while supplies from Angola stood at 0.3 million tonnes. Although Oman is located in West Asia, it has a sizable coastline facing the Arabian Sea and the Gulf of Oman, and doesn’t depend on the Strait of Hormuz for the transit of its cargoes. Last year, the average monthly LNG imports from Nigeria, Oman,

and Angola were 0.1 million tonnes, 0.2 million tonnes, and 0.1 million tonnes, respectively.

In 2025, the average monthly LNG import volumes from Qatar stood at 1 million tonnes, followed by imports from the UAE at 0.3 million tonnes. In March–April, the Strait of Hormuz closure meant that just 0.06 million tonnes came from Qatar and 0.13 million tonnes from the UAE during the two months.

“This (recovery in India’s LNG imports) is notable given Asian spot LNG spiked to peak of \$25/mBtu in late Mar ’26 after Qatar’s production halt and Hormuz closure, before easing to around \$17-18/mBtu in May, still about 70% above pre-crisis (level of) about \$10/mBtu. While South Korea and Japan cut imports, India moved in the opposite di-

● INDIA’S LNG IMPORT (million tonnes)

SUPPLIER	MAY 2026	MONTHLY AVERAGE IN 2025
US	0.9	0.2
Nigeria	0.5	0.1
Oman	0.3	0.2
Angola	0.3	0.1
UAE	0.1	0.3
Qatar	0	1
Others	0.1	0.2

SOURCE: KPLER

rection, absorbing cost premium to keep volumes flowing,” Equirus Securities said in a note.

While high and largely inelastic demand from critical sectors like CGD and fertilisers has ensured that LNG imports remain strong, a few other sectors have added to the demand for LNG due to the prevailing weather and fuel supply situation. “India’s power consumption surged over 11% yoy in May ’26. Peak demand hit an all-time high of over 270 GW. With coal carrying about 70% of load and hydro 10%, Ministry of Power directed all gas-based plants on standby for heatwave shortfall coverage, turning high-cost LNG into a non-negotiable grid necessity,” Equirus said.

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● India-Oman trade

YEAR	EXPORTS (\$BN)	IMPORTS (\$BN)	TRADE BALANCE (\$BN)	TOTAL TRADE
2024-25	4.06	6.54	2.48	10.60
2023-24	4.42	4.52	0.10	8.94
2022-23	4.47	7.91	3.44	12.38
2021-22	3.14	6.84	3.70	9.98
2020-21	2.3	3.08	0.78	5.38

SOURCE: COMMERCE & INDUSTRY MINISTRY

West Asia Conflict: How pact with Oman cushions India’s trade

Ravi Dutta Mishra
New Delhi, June 6

THE INDIA-Oman free trade agreement (FTA), which was signed in December last year, entered into force on June 1, at a time when the sustained blockade of the Strait of Hormuz has begun to translate into wider macroeconomic worry for the country due to high dependence on costly imported fuel, fertilisers, and key industrial input items.

While higher freight costs continue to be a major impediment to trade flows with the West Asia region, the trade deal with Oman could offer a breather for India due to its geographical location, which sits outside the Strait of Hormuz. The country is increasingly seen as a critical gateway to the region if the ongoing US-Iran conflict continues.

Trade experts pointed out that, unlike most Gulf countries, which rely on shipping through the Strait of Hormuz, much of Oman’s coastline is located outside the strait, directly on the Arabian Sea and the Gulf of Oman. This allows major ports such as the Port of Salalah and Port of Duqm to remain accessible even when traffic through the strait is disrupted.

Ties with Oman would be important for India as its trade with the UAE has taken a big hit due to the West Asia crisis. After signing a trade deal with the UAE in 2022, Abu Dhabi emerged as one of the largest trade partners in the Gulf region, with overall trade crossing \$100 billion during the last financial year.

However, overall trade with the UAE has declined by 35% in April. This is primarily because only two UAE ports, Khor Fakkan and Fujairah, have managed to avoid operational difficulties amid the blockade.

Trade data also showed that while commerce with West Asian countries has come under pressure since the Strait of Hor-

muz was closed on March 2, trade with Oman has picked up. Energy imports from Iraq, Qatar and the UAE declined sharply in April, but shipments from Oman jumped 246.42% to \$1.48 billion last month, compared to \$429.58 million in April last year.

Before the war, India’s trade had been heavily dependent on the UAE and Saudi Arabia. The two countries accounted for 8.4% and 2.7% of India’s total exports, respectively, in FY25, while Oman’s share stood at just 0.9%. On the import side, the UAE and Saudi Arabia comprised 8.8% and 4.2% of India’s total imports, with Oman again at 0.9%, Commerce Ministry data showed. Under the deal, Oman has granted zero-duty

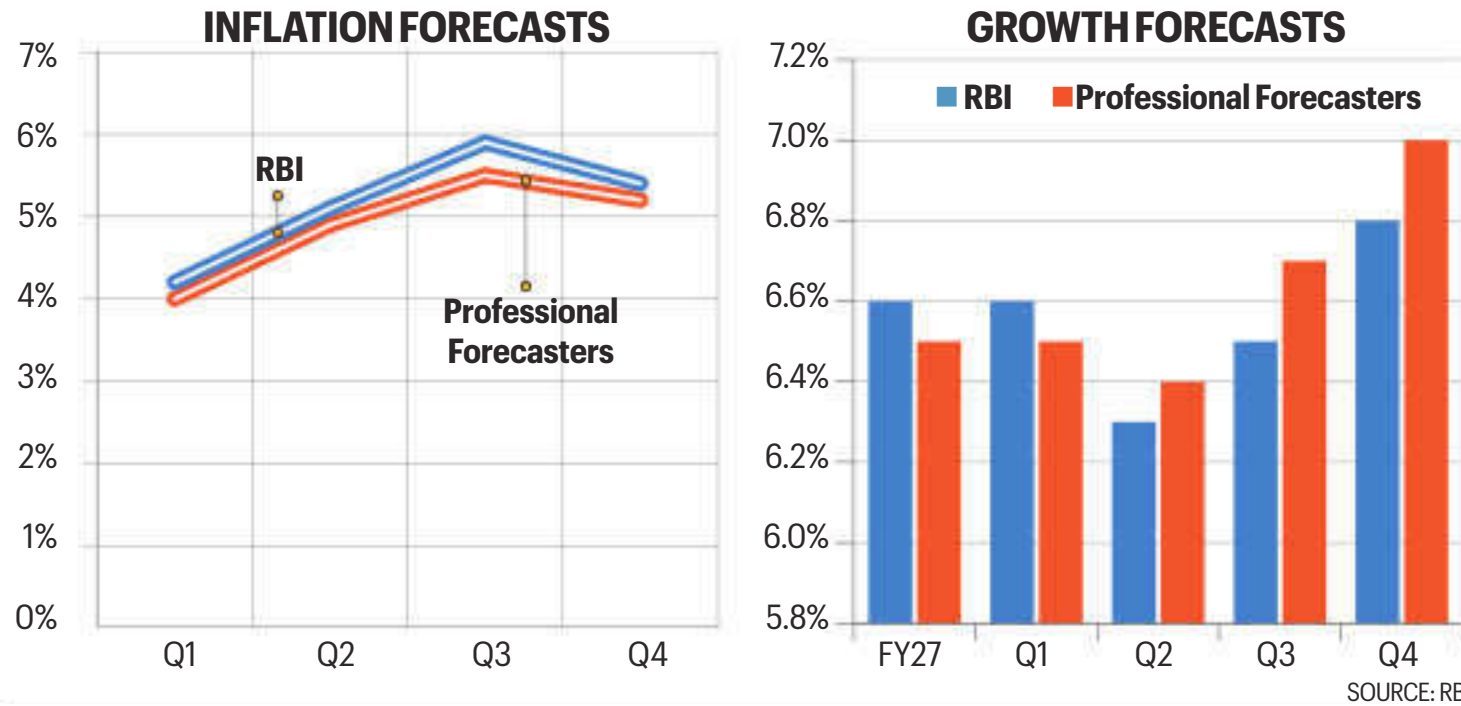
access on 98% of tariff lines covering 99% of India’s exports, boosting prospects for products ranging from petroleum goods to machinery and steel. “The agreement deepens India’s energy security by ensuring access to Omani crude oil, LNG, fertilisers, methanol and ammonia, worth over \$7.2 billion in imports in FY2026,” former trade officer and founder of think tank Global Trade and Research Initiative (GTRI), Ajay Srivastava said.

Although over 80% of Indian exports already entered Oman at relatively low average tariffs of around 5%, duties on certain products reached as high as 100%, he said, adding that their elimination is expected to improve the competitiveness of Indian goods in the Omani market, though export growth will inevitably be constrained by the country’s relatively small market size. The pact with Oman also assumes significance given its role in the ongoing West Asia conflict. Iran’s Deputy Foreign Minister Kazem Gharibabadi has said that Tehran is working on arrangements to determine what services “Iran and Oman are providing in the Strait of Hormuz”.

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● Is the RBI pessimistic on inflation?

With the West Asia war dragging on, the Reserve Bank of India (RBI) on Friday raised its headline retail inflation forecast for 2026-27 by 50 basis points to 5.1%, with the average for October–December 2026 at 5.9% — almost touching the upper bound of the central bank’s 2%-6% mandate. Meanwhile, the median estimate for the same quarter of 40 economists from outside the RBI was a significantly lower 5.5%, as per the central bank’s Survey of Professional Forecasters. Forecasts — by both the RBI and external economists — for the other three quarters of the current financial year were all within 20 bps of each other, although it was the central bank’s numbers which were higher in every instance. Even when it comes to underlying price pressures, the professional forecasters see core inflation — or inflation excluding food and fuel — averaging 4.4% in 2026-27 as against the RBI’s latest projection of 4.7%. On the growth front, views are more closely aligned: 6.6% by the RBI as against 6.5% by the forecasters. But the quarterly numbers revealed the central bank to be more cautious.



• BRIEFLY



Mirra Andreeva defeated Maja Chwalinska in the final. AP

Mirra Andreeva claims her maiden Grand Slam despite jittery start

MIRRA ANDREEVA became the youngest French Open champion in over three decades on Saturday as she swept aside surprise finalist Maja Chwalinska 6-3 6-2 on a blustery Court Philippe Chatrier, where the Pole seemed frozen by the magnitude of the occasion. The 19-year-old Russian, the youngest Roland Garros women's singles winner since Monica Seles won her third straight title in 1992 aged 18, overcame a jittery start to claim her maiden Grand Slam crown, growing in authority as the match wore on while Chwalinska struggled to settle on the big stage. The world number 114, who will rise to 21st in the world rankings on Monday, had captivated Roland Garros with her tactical intelligence and fearless variety in a nine-match winning run that began in qualifying but the magic deserted her in the final despite the support of Polish fans. REUTERS